European Journal of Neuroscience, Vol. 12, pp. 1113–1122, 2000 © European Neuroscience Association

Red nucleus lesions impair overground locomotion in rats: a kinetic analysis

Gillian D. Muir[1] and Ian Q. Whishaw[2]
[1]Department of Veterinary Physiological Sciences, University of Saskatchewan, Saskatoon, Saskatchewan, Canada
[2]Department of Psychology and Neuroscience, University of Lethbridge, Lethbridge, Alberta, Canada

Keywords: excitotoxic lesion, gait, ground reaction forces, kinetic, mechanical energy, potential

Abstract

The red nucleus is a prominent brainstem nucleus in mammals which is thought to be involved in production of skilled limb movements. The presence of the red nucleus and associated rubrospinal tract in animals that do not produce skilled limb movements, however, suggests that these structures might also be involved in control of more general limb actions, such as those occurring during locomotion. The present study investigates this question by measuring the three-dimensional ground reaction forces produced by locomoting rats with unilateral excitotoxic lesions of the red nucleus. Twenty-four to 48 h after the lesion, rats moved with an asymmetric gait during which abnormal braking and propulsive forces were produced during the dual contact time of the forelimb contralateral to the lesion and the ipsilateral hindlimb. Rats did not recover normal symmetrical locomotion within the 55-day duration of the study. The persistent asymmetry produced by red nucleus ablation provides the first unequivocal demonstration that the red nucleus plays a role in ongoing overground locomotion in the rat. Species differences in phylogeny and connectivity of the red nucleus are discussed, as well as the possibility that there is a general compensatory response to unilateral CNS injury in the rat.

Introduction

The red nucleus is prominent brainstem nucleus which has been the subject of much investigation in the past century (for reviews, see Massion, 1967, 1988; Houk, 1991). It has been implicated in control of limb movement, largely because of its major output to the spinal cord, the rubrospinal tract. Recent research has investigated the function of the red nucleus during goal-directed and/or skilled limb movements, especially of the forelimbs (Houk *et al*., 1988; Houk, 1991; Levesque & Fabre-Thorpe, 1990; Whishaw *et al*., 1990, 1992; Whishaw & Gorny, 1996). Theories developed from this research have ranged from general suggestions that the red nucleus is involved in controlling arm musculature (Lawrence & Kuypers, 1968; Miller & Sinkjaer, 1998) and/or providing tonic support for digit movements (Whishaw & Gorny, 1996) to more complex theories of motor learning (Paillard, 1978; Massion, 1988), triggering conditioned motor responses (Schmied *et al*., 1988), or switching functions (Kennedy, 1990).

Emphasis on the role of the red nucleus in skilled movements, however, does not address whether this structure is involved in less skilled, more stereotyped limb movements, such as those occurring in locomotion. The red nucleus and a prominent rubrospinal tract are present in all limbed terrestrial vertebrates, including those which do not use their limbs for skilled movements, suggesting that the red nucleus might play a role during locomotion (Huisman *et al*., 1982; ten Donkelaar, 1982, 1988). While lesion studies have shown that input from the red nucleus is not required for production of the basic limb movements occurring during locomotion (for review, see Armstrong, 1986), these experiments have involved mainly decerebrate preparations and, as such, have not investigated the locomotor role of the red nucleus in awake, behaving animals. Additionally, lesions of the red nucleus were induced electrolytically, a procedure which destroys both cell bodies and adjacent neuronal processes. Results from recording and stimulation studies have provided some support for a role of the red nucleus in locomotion; rubrospinal activity is phasically modulated during locomotion, while stimulation of rubrospinal neurons during treadmill locomotion results in phase-dependent facilitation of both flexor and extensor muscles of the limbs (Arshavsky *et al*., 1988; Rho *et al*., 1999). Indeed, Armstrong (1986) has argued that the red nucleus may be important in correcting and adapting ongoing locomotion and suggests that destroying rubral cells with a neurotoxin and investigating residual locomotor capacity could clarify its role in locomotion. Here, we investigate this question by examining the locomotor behaviour of rats with excitotoxic lesions of the red nucleus.

To quantify the role of individual limbs during locomotion in behaving animals, we measured three-dimensional ground reaction forces (the forces acting on the ground through the limbs) while the animals moved overground unrestrained. Ground reaction forces have been used as a sensitive and noninvasive technique to evaluate the role of each limb in support and propulsion of the body during locomotion in many species (Jayes & Alexander, 1978; Merkens *et al*., 1986; Budsberg *et al*., 1987; Winter, 1990; Clarke, 1995; Lavioe *et al*., 1995; Muir *et al*., 1996, 1998; Bertram *et al*., 1997; Muir & Whishaw, 1999a, b). Ground reaction forces can also provide an indication of overall gait compensation, through analysis of the mechanical energetics of overground locomotion (Cavagna *et al*., 1977; Muir *et al*., 1996). We have previously used ground reaction forces to quantify locomotor symmetry in rats with unilateral CNS lesions (Muir & Whishaw, 1999a, b). Locomotor asymmetry persists after unilateral striatal dopamine depletion, a lesion which produces

Correspondence: Dr G. Muir, as above
E-mail: gillian.muir@usask.ca

Received 7 October 1999, revised 27 November 1999, accepted 6 December 1999

14609568, 2000, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.1046/j.1460-9568.2000.00987.x by Erasmus University Rotterdam Universiteitsbibliotheek, Wiley Online Library on [02/12/2022]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

well documented asymmetries in other behaviours (Schallert *et al*., 1982; Dunnett *et al*., 1986; Robinson *et al*., 1994; Miklyaeva *et al*., 1995; Muir & Whishaw, 1999b). In contrast, rats quickly regain normal symmetrical locomotion after lesions of the pyramidal tract, a structure which plays little to no role during overground locomotion (Muir & Whishaw, 1999a). In the present study, we demonstrate that the red nucleus plays an ongoing role during overground locomotion in the rat.

Materials and methods

Subjects

Twenty-two female Long–Evans rats, weighing 230–300 g were used. The animals were raised and housed in hanging wire cages at the University of Lethbridge vivarium. All rats were food deprived to 90% of *ad lib* body weight by providing once-a-day measured feeding. Twelve rats were unoperated and ten rats underwent left red nucleus ablation. All animals were cared for according to standards outlined by the Canadian Council on Animal Care.

Surgery

Rats were anaesthetized with sodium pentobarbitol (MTC Pharmaceuticals, Ontario, Canada) at a dose of 40 mg/kg, injected intraperitoneally. Red nucleus ablation was induced, as previously described (Whishaw *et al*., 1990), by injection of 0.6 μL of a 10 μg/μL solution of ibotenic acid (ICN Pharmaceuticals, California, USA) dissolved in 0.1 M phosphate-buffered saline, pH 7.4, into the left red nucleus. Coordinates for the injection were, with the skull horizontal, 5.6 mm posterior and 0.7 mm lateral to bregma, and 6.5 mm ventral to the dura. Injections were made over 3 min with 7 min allowed for diffusion. All animals were allowed to recover for at least 24 h before recording of locomotor data.

Histology

After recording of all locomotor data, animals were deeply anaesthetized and perfused intracardially with a solution of 0.9% NaCl and 10% formalin. Brains were removed and placed in a 30% sucrose-formalin solution, and cut in 40 μm frozen sections. Alternate sections were mounted and stained with Cresyl violet.

Locomotor measurements

As previously described, rats were trained to move along the length of a 3 m long × 20 cm wide runway for a food reward (Muir & Whishaw, 1999a, b). A force platform (Muir *et al*., 1996, 1998) was built in level with the runway surface and located equidistant from either end of the runway. The platform measured force in three orthogonal directions: vertical, fore-aft and mediolateral. Animals were videotaped at 120 frames per s (Peak Performance Technologies, Englewood, CO) from a stationary lateral view as they moved along the runway unrestrained. Ground reaction force data was analogue-to-digital converted (RC Electronics, San Raphael, CA) and collected on computer. A pair of infra-red beams, positioned at either end of the platform, were used to calculate average velocity across the platform and also used to synchronize the videotape with ground reaction force data (Muir *et al*., 1996, 1998; Muir & Whishaw, 1999a, b). Ground reaction force data were saved only from those passes in which the rat was observed to maintain a constant speed across the platform, and in which at least two limbs (forelimb followed by the ipsilateral hindlimb) were placed completely on the platform.

Data analysis

The videotape and ground reaction forces for each single pass were classified according to speed, gait and the identity of the limbs which contacted the platform. Typically, the animal contacted the platform with a single forelimb followed separately by the ipsilateral hindlimb, such that each pass would measure forces exerted through either both left limbs or both right limbs. Passes were considered acceptable if the forces in the fore-aft direction demonstrated equal amounts of deceleration and acceleration, indicating that the animal maintained a constant speed over the platform. For each individual animal, the ground reaction force data for 5–10 acceptable passes at each speed were averaged, keeping the data from left and right limbs separate. Averages and standard error of ground reaction forces for each experimental group of animals were calculated from averaged data for individuals. Statistical comparisons were made using analysis of variance (SPSS Inc, Chicago, IL).

To obtain total ground reaction forces acting on the rat throughout the stride, it is necessary to sum the data obtained separately for right and the left limbs. Averaged ground reaction force traces for the left and the right limbs were aligned according to the amount of limb overlap and summed arithmetically. The amount of limb overlap at each speed was obtained by single-frame analysis of videotape, as previously described (Muir & Whishaw, 1999a, b). Briefly, the time of initial limb contact and lift-off for each limb obtained from videotape was used to calculate the support duration for each limb, the overlap between contralateral limb pairs (i.e. between the right and left forelimbs or between the right and left hindlimbs) and the overlap between diagonal limb pairs (i.e. between right forelimb and left hindlimb or between left forelimb and right hindlimb). For each of these variables, averages for each individual were obtained at different speeds and subsequently used to obtain averages for each experimental group.

Averaged ground reaction force records were analysed to yield mechanical (potential and kinetic) energy changes over time using custom written software (Muir *et al*., 1996, 1998). Potential energy changes reflect the vertical position of the body and kinetic energy changes reflect the fore-aft movement of the body. Acceleration in the vertical and fore-aft directions was integrated over time to yield vertical and fore-aft velocity. Vertical velocity was integrated over time to produce vertical displacement. Constants for integrations of both vertical acceleration and vertical velocity were assumed to be zero, a valid assumption when the integration was performed over an integral number of steps. The integration constant for fore-aft acceleration was the average velocity for each group of animals (68 cm/s for control, 62 cm/s for red nucleus-lesioned rats). The kinetic energy of the centre of mass due to velocity in the fore-aft direction was calculated over time as $(mv^2)/2$ where m is body mass in kilograms and v is fore-aft velocity. Potential energy changes were calculated over time as mgh, where g is the gravitational acceleration constant (9.8 m/s^2) and h is the change in vertical displacement of the centre of mass.

Results

Evaluation of lesions

Cell counts in the region of the red nucleus were made at three different rostrocaudal levels in those animals in which ibotenic acid was injected into the left red nucleus, as previously described (Whishaw *et al*., 1990, 1992; Whishaw & Gorny, 1996). In eight of ten animals, at least 95% of the cells in the left red nucleus were ablated (Fig. 1). The results for only these eight animals are reported

14609568, 2000, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.1046/j.1460-9568.2000.00987.x by Erasmus University Rotterdam Universiteitsbibliotheek, Wiley Online Library on [02/12/2022]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

below. In one of the two remaining animals, there was no difference in the number of cells between right and left red nuclei – the injection site was adjacent to, but not directly into, the red nucleus. No locomotor deficits were seen 24 h post-lesion in this animal and ground reaction forces were indistinguishable from unlesioned animals. In the second animal, approximately 80% of the cells in the left red nucleus were ablated. Ground reaction forces in this animal were similar to those of completely lesioned animals.

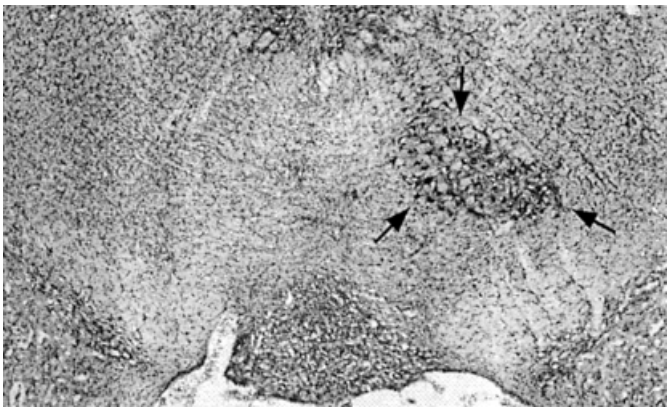

FIG. 1. Photomicrograph of a transverse section through the midbrain showing the extent of red nucleus ablation. Arrows indicate the position of the intact red nucleus on the right. Absence of cellular staining on the left demonstrates that the cells of the left red nucleus were completely ablated in this animal, as in the other eight animals used in this study.

Locomotor analysis

Controls

The most consistent gait adopted by control rats moving along the pathway was a trot, in which each diagonal limb pair (e.g. right fore- and left hindlimb or left fore- and right hindlimb) contacted the ground alternately (see bar diagram, Fig. 3). Rats trotted at medium to fast speeds in this study (50–85 cm/s), speeds at which there is little overlap between diagonal limb pairs.

Ground reaction forces in normal rats are similar to those found in other quadrupedal species and have been described (Muir & Whishaw, 1999a, b). Briefly, a single peak in vertical ground reaction forces was produced for each diagonal limb pair (Figs 2A and 3A). Unlike dogs, cats and horses, which carry up to 60% of their bodyweight on the forelimbs, rats distributed their weight relatively equally between fore- and hindlimbs. Forces in the fore-aft direction are not distributed equally, however; the forelimbs provided mainly

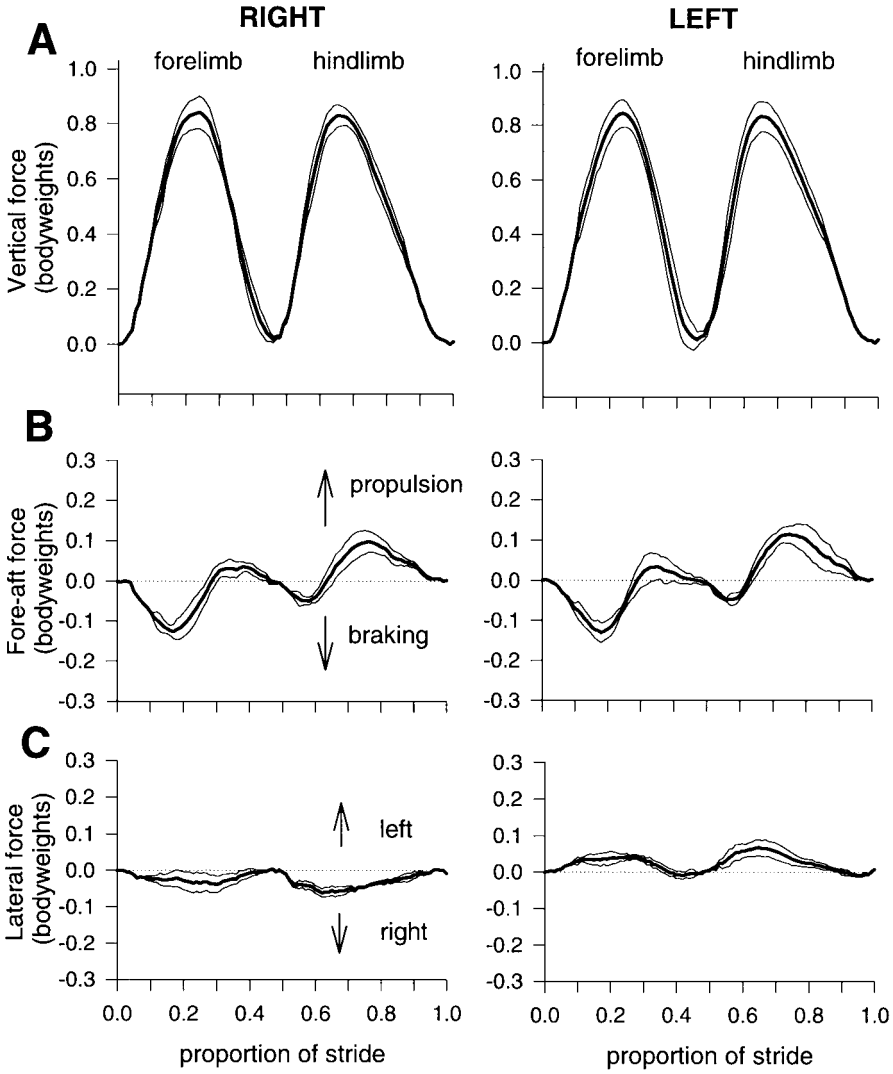

FIG. 2. Ground reaction forces for a single trotting stride plotted against proportion of stride duration for control rats moving at 50–85 cm/s. Data from right and left limbs are shown separately, and forces acting through the forelimb precede those from the hindlimb. Thick line is averaged data from 12 rats, thin lines represent ± one standard error for vertical forces (A), horizontal fore-aft forces (B) and horizontal medio-lateral forces (C). Note the symmetry between right and left sides. (Modified from Muir & Whishaw, 1999a).

14609568, 2000, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.1046/j.1460-9568.2000.00987.x by Erasmus University Rotterdam Universiteitsbibliotheek, Wiley Online Library on [02/12/2022]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

braking forces while the hindlimbs produced propulsive forces (Fig. 2B). Limb timing was such that, when these forces were summed to provide total fore-aft force, a smooth transition from braking force to propulsive force was produced during the stance of each diagonal limb pair (Fig. 3B). Both fore- and hindlimbs produced small laterally directed forces, the hindlimb forces being larger than those produced by the forelimbs (Fig. 2C).

In rats, as in other terrestrial animals, potential and kinetic energy of the body oscillated twice during each stride (Fig. 4; Cavagna *et al.*, 1977). These forms of mechanical energy represent, respectively, the vertical position and the fore-aft velocity of the body. Both forms of energy were lowest during the midstance of each diagonal limb pair and highest during the transfer of weight between diagonal limb pairs. These oscillations occur because, in the first half of the stance phase, the centre of mass falls and slows down as the animal bears weight on the limbs; in the second half of the stance phase, the body reverses direction and is propelled upward and forward. Each cycle is repeated twice for each trotting stride (Fig. 4).

For all control rats, no significant right-left asymmetries could be detected in ground reaction force data, limb timing variables or mechanical energy oscillations.

Red nucleus

Rats with left red nucleus ablation moved with an asymmetric gait which did not return to normal symmetrical locomotion for the duration of the study (55 days). Visually, the gait resembled the trot of control animals, except that the head and forequarters were raised briefly during part of the stride, giving the gait a distinct asymmetry. All animals trotted at medium to fast speeds (50–85 cm/s). Data presented here were recorded from red nucleus-ablated rats at 30 days postoperative. It should be noted that the locomotor asymmetry expressed after ibotenic acid injection into the red nucleus ablation

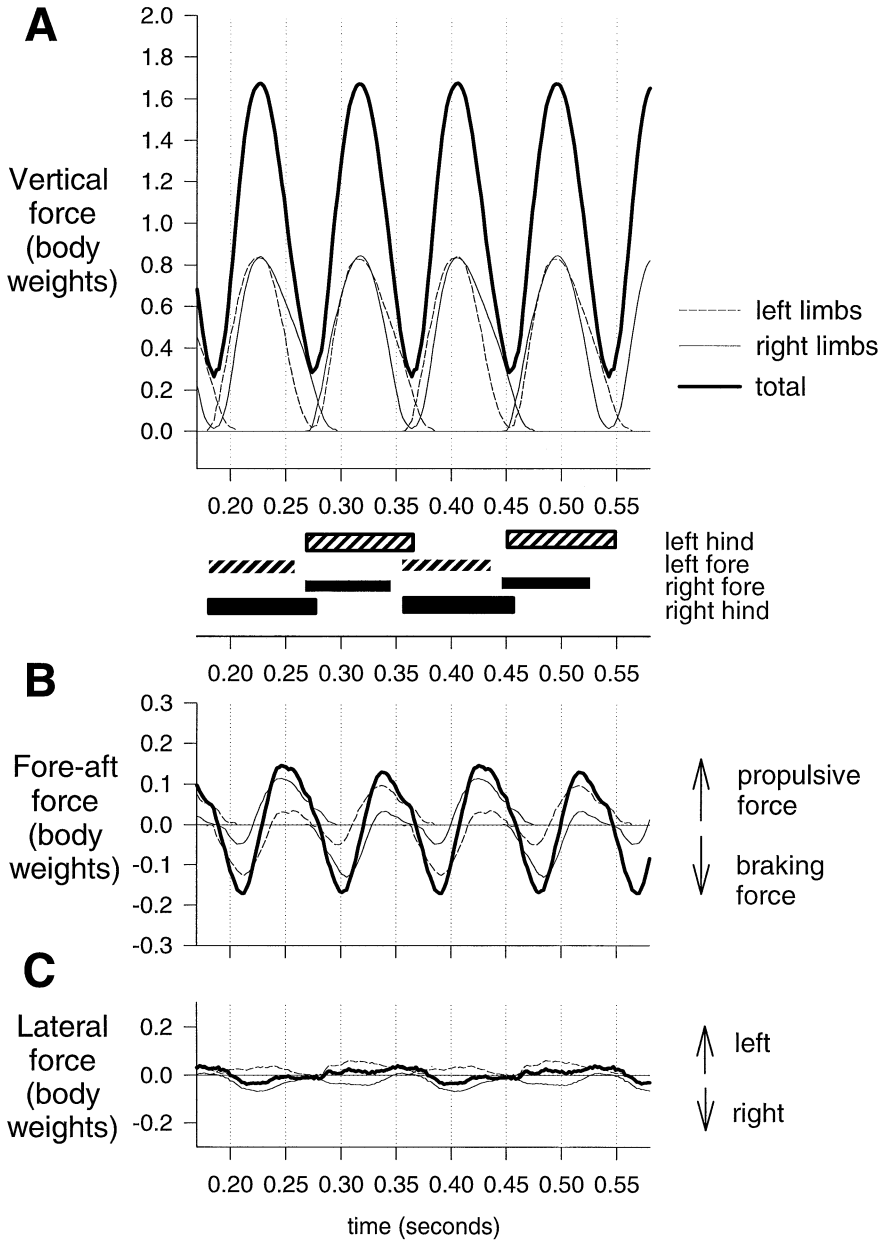

FIG. 3. Summed ground reaction forces for two complete strides plotted against time for control animals trotting at 50–85 cm/s. Thick solid line is total force in the vertical (A), horizontal fore-aft (B) and horizontal medio-lateral (C) directions. Thin solid lines indicate forces produced by the right limbs, thin dashed lines indicate forces produced by the left limbs. Solid bars demonstrate the duration of ground contact for each limb. Notice the even oscillations of total vertical force around bodyweight of the animal. Total fore-aft forces (B) also oscillate evenly between propulsion (in the positive direction) and braking forces (in the negative direction). Modified from Muir & Whishaw, (1999a).

14609568, 2000, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.1046/j.1460-9568.2000.00987.x by Erasmus University Rotterdam Universiteitsbibliotheek, Wiley Online Library on [02/12/2022]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

could potentially be due to a nonspecific effect of injection rather than ablation of red nucleus neurons. While we cannot rule this out, it is less likely in light of the observation that when an injection was made adjacent to the red nucleus and no red nucleus cells were ablated, no locomotor deficits were seen.

There were slight asymmetries in the vertical ground reaction forces, largely due to the reduced support provided by the impaired (right) forelimb. Peak vertical force produced by the impaired forelimb was significantly smaller than that produced by the unimpaired (left) forelimb (Fig. 5A, $P = 0.006$). Single-frame videotape analysis showed that it was during the stance of the impaired forelimb and unimpaired hindlimb that the head and forequarters were raised briefly. In addition, the contact of the unimpaired hindlimb preceded the contact of the impaired forelimb, unlike for control animals in which forelimb contact always preceded hindlimb contact (Fig. 6A). Altogether, these results suggest that the body centre of mass was shifted caudally during the stance of the impaired forelimb and unimpaired hindlimb (Fig. 6A).

The most marked change in ground reaction force occurred in the fore-aft direction. The impaired hindlimb produced a larger peak propulsive force than either the unimpaired hindlimb or the control hindlimb (Figs 2B and 5B; $P < 0.001$). Propulsive impulse (the total propulsive force produced over the stance phase, calculated as the area under the force vs. time curve) was also larger for the impaired hindlimb when compared with the unimpaired hindlimb ($P = 0.047$). No significant differences in peak force or impulse were found between the unimpaired limbs and control data for fore-aft forces. The net fore-aft forces produced by red nucleus-ablated animals, however, was markedly different from control (Fig. 6B). No effective braking or propulsive forces were produced during the support phase of the impaired forelimb and unimpaired hindlimb, largely because the timing of the forces from these limbs was such that they negated each other (Fig. 6B and see below). Normal braking following by propulsion occurred during the stance of the unimpaired forelimb and impaired hindlimb (Fig. 6B).

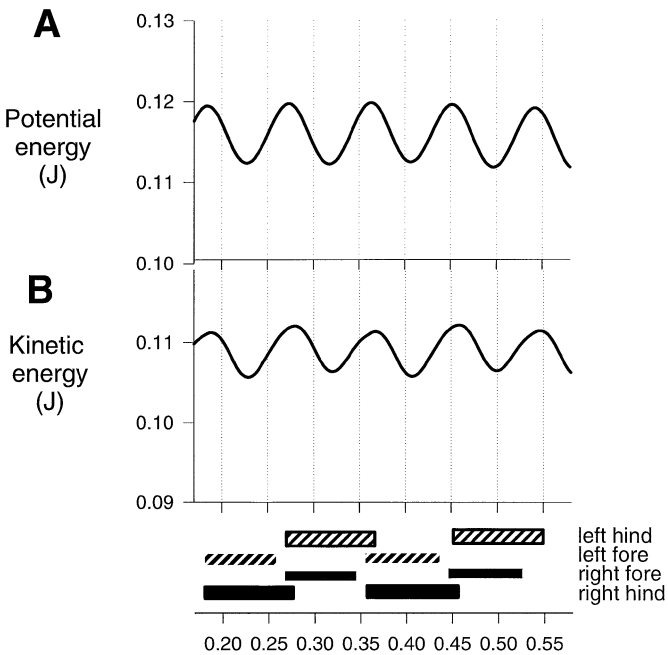

FIG. 4. Potential (A) and kinetic (B) energy oscillations for 2 complete strides plotted against time for control rats trotting at 50–85 cm/s. Solid bars demonstrate the duration of ground contact for each limb. Both potential and kinetic energy oscillated in synchrony, twice during each stride.

Laterally directed forces produced by the unimpaired forelimb were slightly smaller and more medially directed than the forces produced by the impaired forelimb or by the forelimbs of control animals (Fig. 5C; $P < 0.01$). This indicates that lesioned animals were placing the unimpaired forelimb more directly beneath the centre of mass of the body. Lateral force produced by the impaired hindlimb was slightly larger than for control animals but this difference was not statistically significant ($P = 0.063$).

Analysis of ground reaction forces to yield potential and kinetic energy changes within each stride further emphasized the gait asymmetry in red nucleus-lesioned animals (Fig. 7). In lesioned animals, potential energy changes, and thus the vertical movements of the centre of mass, were similar to those occurring in control animals (Fig. 7A). In contrast, kinetic energy changes were distinctly asymmetric in red nucleus-lesioned animals, in that kinetic energy oscillated only once during each stride, rather than twice as in control animals (Fig. 7B). Kinetic energy decreased and then increased during the stance of the unimpaired forelimb and impaired hindlimb but subsequently remained relatively constant throughout the stance phase of the impaired forelimb and unimpaired hindlimb (Fig. 7B). The lack of oscillation during this latter period may be the result of compensatory actions by the unimpaired hindlimb. This limb contacted the ground prematurely compared with control animals, possibly to compensate for decreased function of the impaired forelimb. Propulsive forces from the unimpaired hindlimb were thus produced at the same time as braking forces from the impaired forelimb, reducing net fore-aft force, and thus fluctuations in kinetic energy, to zero (Fig. 6A). In addition, the magnitude of the kinetic energy changes occurring during the stance of the unimpaired forelimb and impaired hindlimb were twice as large as in control animals. This may be an adjustment for the fact that kinetic energy cycled only once during the stride rather than twice as in control animals – the changes that did occur may have needed to be approximately twice as large in lesioned animals in order to move at the same velocity overground as control animals.

Discussion

The present paper provides the first unequivocal demonstration that the red nucleus plays an ongoing role during overground locomotion in the rat. Unilateral lesions of the red nucleus result in a gait characterized by abnormal ground reaction forces during the stance phase of the impaired forelimb and unimpaired hindlimb. This asymmetry causes altered kinetic energy oscillations within each stride, such that kinetic energy cycled only once per stride in lesioned animals rather than twice per stride as in control animals. The fact that this asymmetry does not recover has important implications for understanding the locomotor role of this brainstem structure. Of special interest is the possibility that quadrupedal animals have a general compensatory response to unilateral CNS injury.

Much previous work on red nucleus function has examined the deficits produced during skilled and/or goal-directed movements of the limbs and digits after red nucleus lesions. Less information is available on the role of this structure during more stereotyped movements such as locomotion. Work in the cat has established that the red nucleus is not required for initiation of locomotion, nor for the production of basic locomotor movements of the limbs (for review, see Armstrong, 1986). After red nucleus lesions or transections of the dorsolateral spinal funiculus, including the rubrospinal tract, locomotor deficits during overground movement appeared to be

14609568, 2000, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.1046/j.1460-9568.2000.00987.x by Erasmus University Rotterdam Universiteitsbibliotheek, Wiley Online Library on [02/12/2022]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

limited to failure of the limb to flex sufficiently during the swing phase, so that the paw dragged on the ground or occasionally knuckled over (Armstrong, 1986; Jiang & Drew, 1996). These deficits were attributed to a loss of voluntary control over the distal limb muscles, a function commonly attributed to the red nucleus (Armstrong, 1986; Houk, 1991). In most cases, more severe deficits have been documented during execution of skilled locomotion, such as ladder or beam walking (Armstrong, 1986). This has led to the conclusion that the red nucleus is important largely for correcting and adjusting skilled movements (Armstrong, 1986; Kennedy, 1990; Houk, 1991), although recent work suggests that rubrospinal neurons can influence limb muscle activity during treadmill locomotion (Rho *et al*., 1999).

In contrast to most early findings in the cat, the present results demonstrate that red nucleus neurons in the rat probably play an ongoing role during unperturbed locomotion over a flat surface. There are several differences between previous studies and the present study which may explain the findings presented here. First, many studies have examined deficits after bilateral lesions, which presumably would produce symmetrical changes in the gait which may be difficult to detect. Unilateral lesions produce asymmetries in the locomotor pattern which can be detected relatively easily during gait analysis, by comparing data from right and left sides of the same animal. The use of unilateral lesions is potentially a more sensitive test for assessing whether a particular structure plays a role in a normally symmetrical behaviour such as locomotion. Another methodological difference is the measurement of ground reaction forces in the present study. Measurement of the force placed on the ground by each limb allows a precise and sensitive determination of the compensations that rats make for CNS lesions, by quantifying the role of each limb in support and propulsion.

There are also several important species differences which may also help to explain the differences between results of the present and previous studies. First, the extent of spinal projections of the red nucleus differs between species (Armand, 1982; Huisman *et al*., 1982). For example, the red nucleus projects to all levels of the spinal cord in the rat, but not in the cat, so that red nucleus lesions might be expected to have more extensive effect on quadrupedal locomotion in the rat compared with the cat. In addition, the red nucleus is divided into parvocellular and magnocellular portions in the cat and in primates, including humans. The parvocellular part of the red nucleus receives input from the cerebral cortex and projects to the inferior

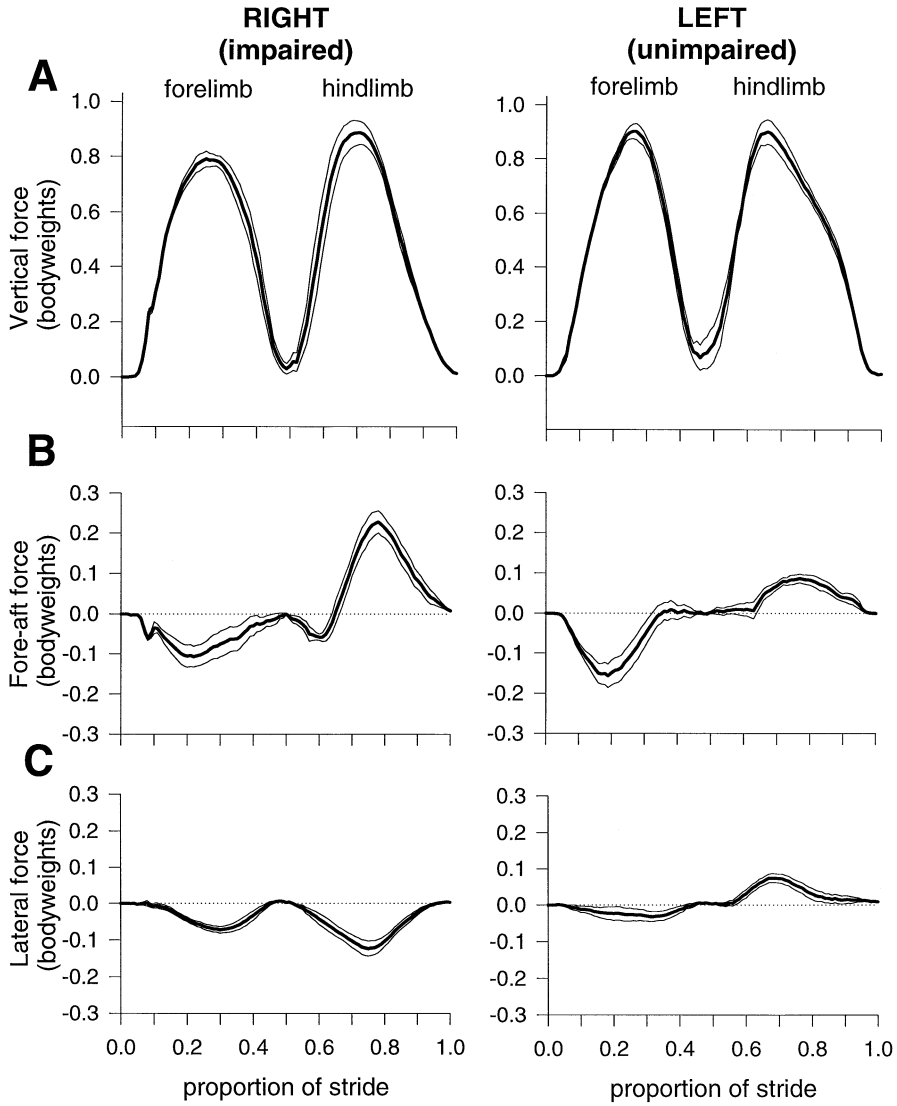

FIG. 5. Ground reaction forces for a single trotting stride plotted against proportion of stride duration for red nucleus-ablated rats moving at 50–85 cm/s. Data from right and left limbs are shown separately, and forces acting through the forelimb precede those from the hindlimb. Thick line is averaged data from eight rats, thin lines represent ± one standard error for vertical forces (A), horizontal fore-aft forces (B) and horizontal medio-lateral forces (C). Note the extended duration of vertical impulse for the left hindlimb (A) and the large propulsive impulse produced by the right hindlimb (B).

14609568, 2000, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.1046/j.1460-9568.2000.00987.x by Erasmus University Rotterdam Universiteitsbibliotheek, Wiley Online Library on [02/12/2022]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

olivary nucleus and thence to the cerebellum; the magnocellular part receives inputs from the dentate and interpositus nuclei in the cerebellum and in turn is the source of the rubrospinal projections (Angaut & Cicirata, 1988; Daniel *et al*., 1988; Giuffrida *et al*., 1988; Jankowska, 1988). In contrast, the distinction between these two portions of the red nucleus is less clear in the rat – red nucleus neurons which project to the inferior olivary nucleus also send collaterals to the spinal cord in the rat (Kennedy, 1990). In addition, cerebellar and cerebral inputs have been shown to converge on same neurons in the red nucleus(Giuffrida *et al*., 1988). Thus, separate functions have been ascribed to the different parts of the red nucleus in species such as the cat and monkey, whereas these roles are inseparable in the rat. Lesions of any part of the red nucleus in the rat would be therefore expected to affect both the cerebellar and the spinal projections more extensively than in other species.

The evolutionary emergence of the rubrospinal tract is consistent with the hypothesis that the red nucleus plays an important role in locomotion. A rubrospinal projection is present in all terrestrial vertebrates, except snakes, and has also been identified in a ray which uses pectoral fins for locomotion (Smeets & Tumerick, 1981; ten Donkelaar, 1982). This strongly suggests that the red nucleus exerts general control over movements of the limbs. Furthermore, a gradual

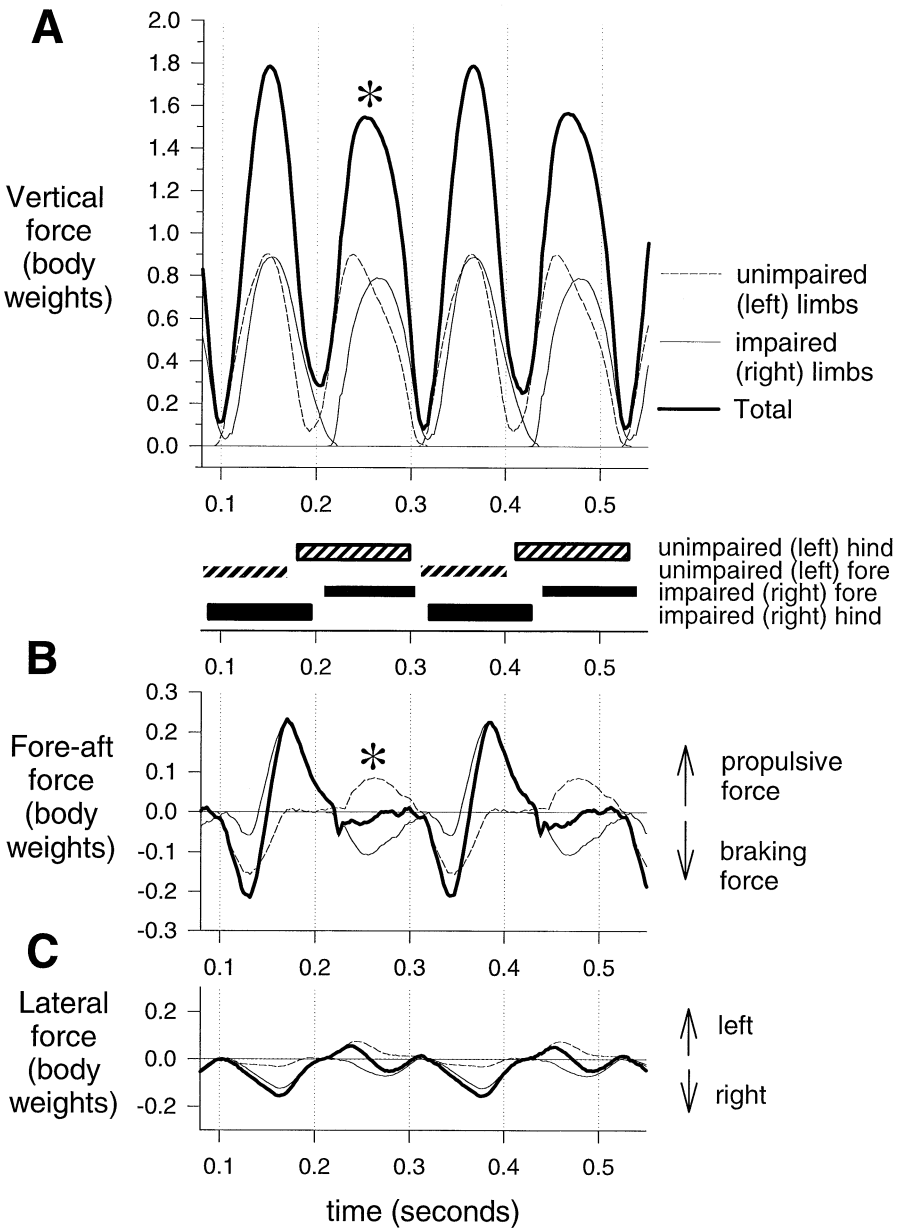

FIG. 6. Summed ground reaction forces for two complete strides plotted against time for red nucleus-ablated animals trotting at 50–85 cm/s. Thick solid line is total force in the vertical (A), horizontal fore-aft (B) and horizontal medio-lateral (C) directions. Thin solid lines indicate forces produced by the right limbs, thin dashed lines indicate forces produced by the left limbs. Solid bars demonstrate the duration of ground contact for each limb. Notice the uneven oscillations of total vertical force around bodyweight within each stride, and the asynchrony of the peak vertical forces from the right (solid line) and left (dashed line) limbs during contact of the left hindlimb and right forelimb (asterisk in A). Fore-aft forces also oscillate unevenly due to the lack of net force-aft force (asterisk in B) during the contact of left hindlimb and right forelimb. Note also that medio-lateral forces (C) are larger than for control animals.

14609568, 2000, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.1046/j.1460-9568.2000.00987.x by Erasmus University Rotterdam Universiteitsbibliotheek, Wiley Online Library on [02/12/2022]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

decrease in collateralization of the rubrospinal projections to different levels of the spinal cord emerges with phylogeny, both across different animal Classes (i.e. from reptiles to birds to mammals) and within mammalian Orders (from opposums to rats to cats to monkeys). Reduced collateralization of rubrospinal pathways is associated with enhanced somatotopic organization of these projections and, behaviourally, an increasing facility for independent movements of the limbs (Huisman *et al*., 1982). Thus a red nucleus lesion would be expected to affect locomotion less severely in the rat compared with the opossum, but more severely in the rat compared with the cat.

What role does the red nucleus play during locomotion in the rat? As previously mentioned, much research has examined the role of the red nucleus during skilled and/or goal-directed movements of the forelimbs. During steady overground locomotion, however, limb movements are relatively stereotyped and many aspects of the basic locomotor synergy are determined at the spinal level. Animals with a complete spinal transection can support their weight and step on a moving treadmill over a range of speeds, with limb kinematics and muscle activation patterns indistinguishable from those of intact animals (Barbeau & Rossignol, 1987; Lovely *et al*., 1990). For example, the transition between stance and swing phases, which largely determines the duration of stance, is produced by a reduction of afferent input as extensor muscles are unloaded near the end of stance (Duysens & Pearson, 1980; Conway *et al*., 1987). Therefore, aspects of limb movements such as (i) the duration of the movement (ii) aiming of the limb, and (iii) force development in limb muscles, which have all been suggested to be related to rubral activity during skilled forelimb movements (Miller & Sinkjaer, 1998), are perhaps less relevant for studies of rubral function during locomotion. Of course, this does not eliminate the possibility that the red nucleus is normally involved in some or all of these aspects during locomotion in intact animals (see Rho *et al*., 1999), but does mean that input from the red nucleus is not required.

In contrast, evidence from the present study reveals that there is some aspect of overground locomotion that requires red nucleus activity, at least in the rat. It is possible, for example, that overground locomotion, unlike treadmill locomotion, requires continual step-to-step adjustments as the animal moves over the substrate. These adjustments are potentially mediated by the cerebellum and ablation of the red nucleus removes an important source of input to the cerebellum, via the olivary nucleus, as well as destroying the rubral output to the spinal cord. This is especially true in the rat, where the rubrospinal and rubro-olivary projections arise from the same cells. Reducing the cerebellar influence over one side of the body will thus interfere with ongoing step-by-step control. The precise nature of the role of the red nucleus is difficult to determine, however, in part because of the behavioural compensations during locomotion that may mask some of the missing function (see below).

The nature of the gait compensation for red nucleus ablation suggests that (i) the forelimb is more influenced by red nucleus input than is the hindlimb and/or (ii) rats adopt a general locomotor strategy for unilateral CNS injury. The first possibility is implicated by the fact that the ground reaction forces and mechanical energy changes produced by lesioned animals are most abnormal during the contact of the impaired forelimb and essentially normal during impaired hindlimb stance. Anatomical evidence supports this possibility; even though red nucleus neurons extend projections to the entire length of the spinal cord in rats, it is known that only 20% of the rubrospinal fibres extend to the lumbosacral enlargement (Huisman *et al*., 1982). Another, although not exclusive, possibility is that the forelimb is performing a more complicated task during locomotion and thus is more impaired by any CNS lesion. The present results would also suggest that altered reaching movements seen in rats after red nucleus ablation may relate more to general control over limb musculature rather than a specific effect on forelimb prehension mechanisms (Whishaw *et al*., 1992; Whishaw & Gorny, 1996).

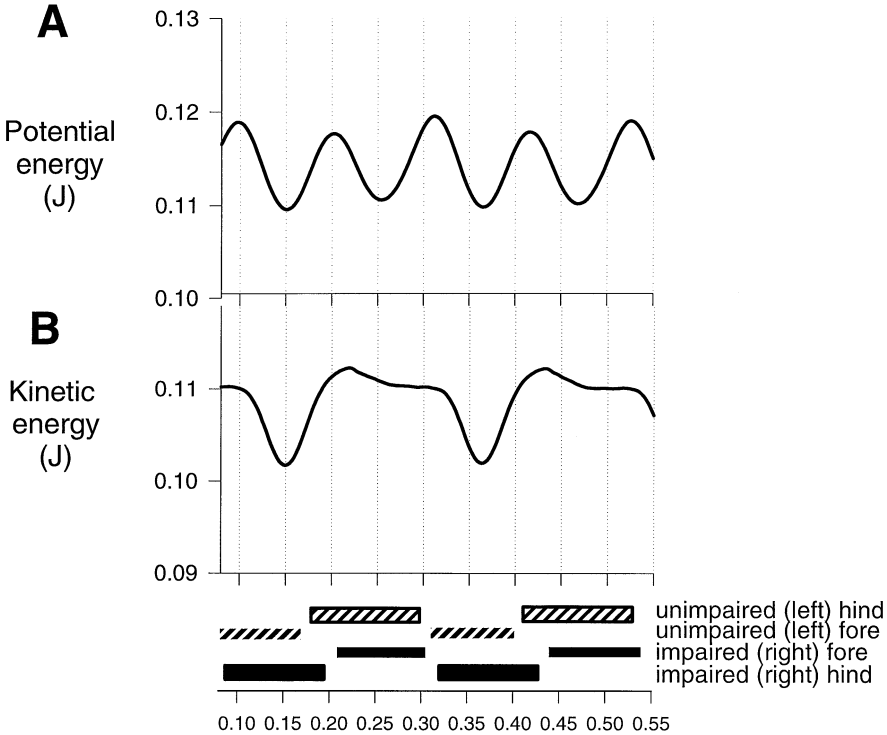

FIG. 7. Potential (A) and kinetic (B) energy oscillations for two complete strides plotted against time for red-nucleus ablated rats trotting at 50–85 cm/s. Solid bars demonstrate the duration of ground contact for each limb. Potential energy oscillates twice each stride as in control animals. In contrast, kinetic energy undergoes only one cycle per stride, remaining relatively constant during the stance of the impaired forelimb and unimpaired hindlimb.

14609568, 2000, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.1046/j.1460-9568.2000.00987.x by Erasmus University Rotterdam Universiteitsbibliotheek, Wiley Online Library on [02/12/2022]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

The gait adopted by rats with red nucleus ablation could also be, in part, a general compensatory strategy for unilateral CNS injury. Rats with unilateral striatal dopamine depletion or immediately after unilateral pyramidal tract transection also produce abnormal ground reaction forces, and abnormal mechanical energy oscillations, during limb contact of the impaired forelimb and unimpaired hindlimb (Muir & Whishaw, 1999a, b). These similarities suggest that there are aspects of the gait pattern presented here that are not necessarily directly related to the pathology in question, but could be more relevant to the compensatory response of the animal. The nature of this response might be, in part, a result of constraints imposed on terrestrial forms of locomotion. It is known, for example, that terrestrial locomotion is energetically costly compared with other forms of locomotion such as flying and swimming (Schmidt-Nielsen, 1972). In order to reduce the cost of overground movement, terrestrial animals have developed mechanisms that involve regular oscillations in different forms of energy, including potential and kinetic energy, throughout the stride (Cavagna *et al*., 1977). These energy oscillations are altered by CNS lesions which, in turn, probably alter (i.e. increase) the cost of terrestrial locomotion. Measurement of the mechanical and energetic changes occurring during overground locomotion may be an insightful method for investigating locomotor adaptations after CNS injury, one that has hitherto been unexplored.

In summary, unilateral red nucleus ablation in rats produces a persistent asymmetry in the ground reaction forces and mechanical energy changes occurring during unrestrained locomotion. This asymmetry suggests that the red nucleus is performing a role during overground locomotion which cannot be compensated for by remaining structures. The nature of the locomotor asymmetry suggests that the forelimb may be more influenced by the red nucleus than the hindlimb, but an equally likely, though not exclusive, explanation is that rats produce a general compensatory response to unilateral CNS injury.

Acknowledgements

This work was supported by the Medical Research Council of Canada and the Natural Sciences and Engineering Research Council of Canada.

References

Angaut, P. & Cicirata, F. (1988) The dentatorubral projection in the rat: an autoradiographic study. *Behav. Brain Res.*, **28**, 71–74.

Arshavsky, Y.I., Orlovsky, G.N. & Perret, C. (1988) Activity of rubrospinal neurons during locomotion and scratching in the cat. *Behav. Brain Res.*, **28**, 193–199.

Armand, J. (1982) The origin, course and terminations of corticospinal fibres in various mammals. *Prog. Brain Res.*, **57**, 329–360.

Armstrong, D.M. (1986) Supraspinal contributions to initiation and control of locomotion in the cat. *Prog. Neurobiol.*, **26**, 273–361.

Barbeau, H. & Rossignol, S. (1987) Recovery of locomotion after chronic spinalization in the adult cat. *Brain Res.*, **412**, 84–95.

Bertram, J.E.A., Lee, D.V., Todhunter, R.J., Foels, W.S., Williams, A.J. & Lust, G. (1997) Multiple force platform analysis of the canine trot: a new approach to assessing basic characteristics of locomotion. *Vet. Comp. Orthop. Traumatol.*, **10**, 160–169.

Budsberg, S.C., Verstraete, M.C. & Soutas-Little, R.W. (1987) Force plate analysis of the walking gait in healthy dogs. *Am. J. Vet. Res.*, **48**, 915–918.

Cavagna, G.A., Heglund, N.C. & Taylor, C.R. (1977) Mechanical work in terrestrial locomotion: two basic mechanisms for minimizing energy expenditure. *Am. J. Physiol.*, **233**, R243–R261.

Clarke, K.A. (1995) Differential fore- and hindpaw force transmission in the walking rat. *Physiol. Behav.*, **58**, 415–419.

Conway, B.A., Hultborn, H. & Kiehn, O. (1987) Proprioceptive input resets central locomotor rhythm in the spinal cat. *Exp. Brain Res.*, **68**, 643–656.

Dunnett, S.B., Whishaw, I.Q., Jones, G.H. & Isacson, O. (1986) Effects of dopamine-rich grafts on conditioned rotation in rats with unilateral 6-hydroxydopamine lesions. *Neurosci. Lett.*, **68**, 127–133.

Duysens, J.D. & Pearson, K.G. (1980) Inhibition of flexor burst generation by loading ankle extensor muscles in walking cats. *Brain Res.*, **187**, 321–332.

Daniel, H., Angaut, P., Batini, C. & Billard, J.M. (1988) Topographic organization of the interpositorubral connections in the rat: a WGA-HRP study. *Behav. Brain Res.*, **28**, 69–70.

Giuffrida, R., Li Volis, G.av & Perciavalle, V. (1988) Influences of cerebral cortex and cerebellum on the red nucleus of the rat. *Behav. Brain Res.*, **28**, 109–111.

Houk, J.C., Gibson, A.R., Harvey, C.F., Kennedy, P.R. & van Kan, P.L.E. (1988) Activity of primate magocellular red nucleus related to hand and finger movements. *Beh. Brain Res.*, **28**, 201–206.

Houk, J.H. (1991) Red nucleus: role in motor control. *Curr. Opin. Neurobiol.*, **1**, 610–615.

Huisman, A.M., Kuypers, H.G.J.M. & Verburgh, C.A. (1982) Differences in collateralization of the descending spinal pathways from red nucleus and other brain stem cell groups in cat and monkey. *Prog. Brain Res.*, **57**, 185–218.

Jiang, W. & Drew, T. (1996) Effects of bilateral lesions of the dorsolateral funiculi and dorsal columns at the level of the the low thoracic spinal cord on the control of locomotion in the cat. *J. Neurophysiol.*, **76**, 849–866.

Jankowska, E. (1988) Target cells of rubrospinal tract fibres within the lumbar spinal cord. *Behav. Brain Res.*, **28**, 91–96.

Jayes, A.S. & Alexander, R.M. (1978) Mechanics of locomotion of dogs (Canis familiaris) and sheep (Ovis aries). *J. Zool. Lond.*, **185**, 289–308.

Kennedy, P.R. (1990) Corticospinal, rubrospinal and rubro-olivary projections: a unifying hypothesis. *Trends Neurosci.*, **13**, 474–479.

Lavioe, S., McFadyen, B. & Drew, T. (1995) A kinematic and kinetic analysis of locomotion during voluntary gait modification in the cat. *Exp. Brain Res.*, **106**, 39–56.

Lawrence, D.G. & Kuypers, H.G.J.M. (1968) The functional organization of the motor system in the monkey II. The effects of lesions of the descending brain-stem pathways. *Brain*, **91**, 15–36.

Levesque, F. & Fabre-Thorpe, M. (1990) Motor deficit induced by red nucleus lesion: reappraisal using kainic acid destructions. *Exp. Brain Res.*, **81**, 191–198.

Lovely, R.G., Gregor, R.J., Roy, R.R. & Edgerton, V.R. (1990) Weight-bearing hindlimb stepping in treadmill-exercised adult spinal cats. *Brain Res.*, **514**, 206–218.

Massion, J. (1967) The mammalian red nucleus. *Physiol. Rev.*, **47**, 383–436.

Massion, J. (1988) Red nucleus: past and future. *Behav. Brain Res.*, **28**, 1–8.

Merkens, H.W., Schamhardt, H.C., Hartman, W. & Kersjes, A.W. (1986) Ground reaction force patterns of Dutch Warmblood horses at normal walk. *Equine Vet. J.*, **18**, 207–214.

Miklyaeva, E.I., Martens, D.J. & Whishaw, I.Q. (1995) Impairments and compensatory adjustments in spontaneous movement after unilateral dopamine depletion in rats. *Brain Res.*, **68**, 23–40.

Miller, L.E. & Sinkjaer, T. (1998) Primate red nucleus discharge encodes the dynamics of limb muscle activity. *J. Neurophysiol.*, **80**, 59–70.

Muir, G.D., Gosline, J.M. & Steeves, J.D. (1996) Ontogeny of bipedal locomotion: walking and running in the chick. *J. Physiol. (Lond)*, **493**, 589–601.

Muir, G.D., Katz, S.L., Gosline, J.M. & Steeves, J.D. (1998) Asymmetric bipedal locomotion – an adaptive response to incomplete spinal injury in the chick. *Exp. Brain Res.*, **122**, 275–282.

Muir, G.D. & Whishaw, I.Q. (1999a) Complete locomotor recovery following corticospinal tract lesions: measurement of ground reaction forces during overground locomotion in rats. *Behav. Brain Res.*, **103**, 45–53.

Muir, G.D. & Whishaw, I.Q. (1999b) Ground reaction forces in locomoting hemi-Parkinsonian rats: a definitive test for impairments and compensations. *Exp. Brain Res.*, **126**, 307–314.

Paillard, J. (1978) The pyramidal tract: two million fibres in search of function. *J. Physiol. (Paris)*, **74**, 155–167.

Rho, M.-J., Lavoie, S. & Drew, T. (1999) Effects of red nucleus microstimulation on the locomotor pattern and timing in the intact cat: a comparison with motor cortex. *J. Neurophysiol.*, **81**, 2297–2315.

Robinson, T.E., Noordhoorn, M., Chan, E.M., Mocsary, Z., Camp, D. & Whishaw, I.Q. (1994) Relationship between asymmetries in striatal dopamine release and the direction of amphetamine-induced rotation during the first week following a unilateral 6-OHDA lesion of the substantia nigra. *Synapse*, **17**, 16–25.

14609568, 2000, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.1046/j.1460-9568.2000.00987.x by Erasmus University Rotterdam Universiteitsbibliotheek, Wiley Online Library on [02/12/2022]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Schallert, T., Upchurch, M., Lobaugh, N., Farrar, S.B., Spirduso, W.W. & Wilcox, R.E. (1982) Tactile extinction: distinguishing between sensorimotor and motor asymmetries in rats with unilateral nigrostriatal damage. *Pharmacol. Biochem. Behav.*, **16**, 455–462.

Schmidt-Nielsen, K. (1972) Locomotion: energy cost of swimming, flying and running. *Science*, **177**, 222–228.

Schmied, A., Amalric, M., Dormont, F.F., Conde, J. & Farin, D. (1988) Participation of the red nucleus in motor initiation: unit recording and cooling in cats. *Behav. Brain Res.*, **28**, 207–216.

Smeets, W.J.A.J. & Tumerick, S.J.B. (1981) Cells of origin of pathways descending to the spinal cord in two chondrichthyans, the shark *Scyliorrhinus canicula* and the ray *Raja clavata*. *J. Comp. Neurol.*, **202**, 473–491.

ten Donkelaar, H.J. (1982) Organization of descending pathways to the spinal cord in amphibians and reptiles. *Prog. Brain Res.*, **57**, 25–68.

ten Donkelaar, H.J. (1988) Evolution of the red nucleus and rubrospinal tract. *Behav. Brain Res.*, **28**, 9–20.

Whishaw, I.Q. & Gorny, B. (1996) Does the red nucleus provide tonic support against which fractionated movements occur? A study of forepaw movements used in skilled reaching by the rat. *Behav. Brain Res.*, **74**, 79–90.

Whishaw, I.Q., Pellis, S.M. & Pellis, V.C. (1992) A behavioral study of the contributions of cells and fibers of passage in the red nucleus of the rat to postural righting, skilled movements, and learning. *Behav. Brain Res.*, **52**, 29–44.

Whishaw, I.Q., Tomie, J. & Ladowsky, R.L. (1990) Red nucleus lesions do not affect limb preference or use, but exacerbate the effects of motor cortex lesions on grasping in the rat. *Behav. Brain Res.*, **40**, 131–144.

Winter, D.A. (1990). *Biomechanics and Motor Control. of Human Movement*. Wiley and Sons, New York, NY.

14609568, 2000, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.1046/j.1460-9568.2000.00987.x by Erasmus University Rotterdam Universiteitsbibliotheek, Wiley Online Library on [02/12/2022]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License